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Rewriting Formulas

*(r = speed)
or $d = rt$*

The distance formula ($d = rt$) states that distance = rate $\times$ time.

Example 1[Hide](#)

Bill drives at an average rate of 70 mph for 2 hours. How many miles does he travel?

$$\begin{aligned} d &= rt \\ &= 70 \times 2 \\ &= 140 \text{ miles} \end{aligned}$$

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You can use algebra to solve for a different variable.

Example 2[Hide](#)

Bill drove 140 miles in 2 hours. What was his average rate of speed in miles per hour?

$$\begin{aligned} d &= rt \\ 140 &= r \times 2 \end{aligned}$$

To isolate r , divide both sides by 2.

$$70 = r$$

You can also use algebra to rewrite the formula so that the formula solves for the rate.

$$d = rt$$

To isolate r , divide both sides by t .

$$\frac{d}{t} = r$$

The new formula solves for rate: $r = \frac{d}{t}$

or $r = \frac{d}{t}$

The two formulas $d = rt$ and $r = \frac{d}{t}$ are equivalent formulas.

The same principles and properties that allow you to solve equations can be applied to rewriting formulas.

To write an equivalent formula, use inverse operations to isolate a variable.

Example 3[Hide](#)



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**Rewriting Formulas**

numbers. Rewrite the formula to solve for x , one of the two

numbers.

$$A = \frac{x+y}{2}$$

Step 1

Multiply both sides by 2.

$$A = \frac{x+y}{2}$$

$$2A = x + y$$

Step 2

Subtract y from both sides.

$$2A - y = x$$

Step 3

Switch the sides of the equation.

$$x = 2A - y$$

The formula $x = 2A - y$ can be used to find a missing number when the average and one of the numbers are known.

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